MVP Hardware Build

Owner	@Alastair Tooth
Document status	FOR REVIEW
Design Link	TBC

- Reference Material
- Hardware
  - General
  - Certification
  - Device Control
  - Casing
  - Display
  - Data Inputs (Essential)
  - Data Inputs (Non-Essential)
- Questions

Reference Material

Hardware must be able to provide functionality matching the below user manual for Industry Competitor MotoGadget, while catering to a high quality User Interface



motoscope_mi… ons.pdf
12 Jan 2024, 01:28 PM

Hardware

Sub Topic	Acceptance Criteria	Business Requirement	Comments / Screenshots

General	• Device should be able to run off a 12V battery powered wiring loom, without causing drain • Device will be mounted to a motorcycles front triple-tree, so must be able to withstand elemental effect ◦ Device must be able to withstand -20 & +80 ° Centigrade temperatures	**Must Have**	
Certification	• Device must meet AS/NZS 3820 Electrical Safety Rating • Device must be built to pass IP67 rating • Device should be IK certified for impact protection ◦ Protection level is to prevent breakage from debris flung up from the road, like pebbles or sticks ◦ **Required IK Impact Protection TBC**		ℹ CE Ratings are not applicable in the Australian Market, so will not be required until product is to be marketed in the EU All electrical equipment imported, and sold in, Australia must comply with AS/NZS 3820 which is the general electrical safety standard
Device Control	• Must be controllable by a bar-mounted control • Control must have 4 Axis pad and a central button • Button function will be: ◦ D-UP: Directional Pad Up ◦ D-DOWN: Directional Pad Down ◦ D-LEFT: Directional Pad Left ◦ D-RIGHT: Directional Pad Right ◦ SELECT: Centre Button	**Must Have**	 Example:
Casing	• CNC Machined Billet Aluminium Case • Design aligned with the following example, but in a rectangular shape: ◦ Example: Motogadget Motoscope Pro	**Must Have**	
Display	• **Display Shape:** ◦ 5” Rectangular Display ◦ Example with similar dimensions: ▪ Freenove 5 Inch Touchscreen Monitor for Raspberry Pi, 800x480 Pixel IPS Display, 5-Point Touch Capacitive Screen, Driver-Free Display Port ◦ Screen should be Bezel-less - display should run to the edge with no border • **Graphics Quality:** ◦ No graphical loss between design and end product ◦ Little to no lag between back end outputs and representation on screen, with no freezing or stuttering	**Must Have**	ℹ Article on determining display quality requirements

	• **Contrast:** Must have high contrast ratio to account for use when in direct sunlight		
Data Inputs (Essential)	**Speedometer** • Unit will be sold with a Hall Sensor ◦ Hall sensor will be attached to the front wheel, and be wired directly into the Unit • Speed will be calculated by determining: ◦ Placement of Hall Sensor from center of wheel to determine distance travelled ◦ Number of rotations of front wheel within a time frame to determine translation to km/h or mph **Tachometer** • Measured via in-built tachometer readout on the bike • Understanding how Tachometers measure revolutions via pulse can be seen via the below links: ◦ MVP Software Development | Tachometer Sensor Reading **Alerts** • Following Alerts must be available on the bike: ◦ High-Beam: Whether high beam is on or not ◦ Neutral Gear: Whether bike is in neutral or not ◦ Engine Temperature: Whether bike's engine is too hot ◦ Indicators: Whether bikes indicator is flashing or not • Each alert comes from individual wires within the motorcycles loom • Each sensor reading is on/off, based on the completion of a circuit within the loom	**Must Have**	**Note: Alternate Speedometer Option** Alternate solution to Hall Sensor is to utilise a GPS to measure speed that can be discussed. Pro's • No installation of wiring required • Similar Solution is now used in many cars Con's • Cost of GPS • Accuracy of GPS (especially in tunnels) GPS is simpler and preferred, so if possible to use then let's discuss.
Data Inputs (Non-Essential)	The following are common sensors on bikes, that it would be good to obtain readouts from and present on the screen. These are **NOT** essential for MVP, so may be taken out of scope where complexities arise. **Engine Temperature (Gauge)** • Engine Temp Alert: ◦ Most motorcycle have an Engine Temperature alert ◦ This uses a single sensor switch that changes from OFF to ON once a certain	**Could Have**	Fuel Gauge, Voltmeter, & Gears should be modular in usage. If a user doesn't have sensor they can unplug the connector

temperature is reached
    - It is legally required to have this alert
- Engine Temp Gauge:
    - Some motorcycles are able to measure the exact temperature of the engine
    - This uses a thermometer sensor that provides the temperature
    - It is **not** legally required to have these, and so can be removed from deliverables where complexity exists

Fuel Gauge

- Used to measure volume of fuel in the tank
- Measured via in-built fuel sensor
- Not all bikes have this sensor, so feature is optional
- Example: GAUGE, FUEL

Gears

- Be able to connect to gear sensor that advises which gear the bike is currently in:
    - All bikes have a sensor advising if the bike is in Neutral, which is why it appears in the Alerts above
    - Not all bikes provide a sensor for other gears, so it is optional
- Example: Gear Position - Shift Indicator Switch | Suzuki GS 850 1981 1734 | Wemoto UK

Voltmeter

- Voltmeters are commonly installed on motorcycles
- These provide the current voltage of the battery, so users are able to understand their batteries charge / health
- Not all bikes have this sensor, so it is optional

Questions

1	Question	Answer
2		
3		
4		
5		
6		